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Symbolic Analysis

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Introduction

One of the key strengths of `EdgeBasedModels.jl` is that all model equations are generated symbolically using [Symbolics.jl](#) and [ModelingToolkit.jl](#). This means we can inspect, simplify, and analyse model equations before running any simulation.

This vignette demonstrates:

- Computing R_0 symbolically for SIR, SEIR, and multi-stage models
- Evaluating symbolic expressions numerically
- Inspecting the ODE equations generated by model builders
- Sensitivity analysis of R_0 to model parameters
- Working with probability generating functions (PGFs) symbolically

Setup

```
using EdgeBasedModels
using ModelingToolkit
using Symbolics
using Plots
```

Symbolic R_0 for SIR

The basic reproduction number R_0 for an edge-based model on a configuration-model network is:

$$R_0 = T \cdot \frac{\psi''(1)}{\psi'(1)}$$

where T is the transmissibility (probability of transmission across a single edge before recovery) and $\psi''(1)/\psi'(1)$ is the excess degree ratio — the expected number of additional edges a neighbour has beyond the one connecting it to the focal node.

For a single-stage SIR with transmission rate β and recovery rate γ , the transmissibility is $T = \beta/(\beta + \gamma)$. For a Poisson network with mean degree κ , the excess degree ratio equals κ , giving:

$$R_0 = \frac{\beta\kappa}{\beta + \gamma}$$

Let us compute this symbolically with `basic_reproduction_number`:

```
@parameters β γ κ
pgf = poisson_pgf(κ)

sir_prog = DiseaseProgression(
  [DiseaseStage(:I; transmission_rate=β), DiseaseStage(:R)],
  [DiseaseTransition(:I, :R, γ)];
  entry=:I
)
scm = StaticConfigurationModel(pgf, sir_prog)
R0_sir = basic_reproduction_number(scm); nothing
```

```
println("Symbolic R0 (SIR): ", R0_sir)
```

Symbolic R_0 (SIR): $(\beta*\kappa) / (\beta + \gamma)$

The result is a symbolic expression in terms of β , γ , and κ that can be manipulated, simplified, or evaluated numerically.

Numeric R_0 Evaluation

We evaluate R_0 at the universal anchor ($\gamma = 0.25$, $R_0 = 2$) by deriving the Poisson configuration-model transmission rate for $\kappa = 5$:

```
# Universal anchors: γ=0.25, R0=2, β derived per scenario (see plan.md)
γ_anchor = 0.25
R0_anchor = 2.0
κ_anchor = 5.0
T_anchor = R0_anchor / κ_anchor
β_anchor = T_anchor * γ_anchor / (1 - T_anchor)
```

```

R0_val = Symbolics.value(
    Symbolics.substitute(R0_sir, Dict(β ⇒ β_anchor, γ ⇒ γ_anchor, κ ⇒ κ_anchor))
)
println("R₀ = ", R0_val)
println("Expected anchor R₀ = ", R0_anchor)

```

```

R₀ = 2.0000000000000004
Expected anchor R₀ = 2.0

```

With $\beta = 1/6$, $\gamma = 0.25$, and $\kappa = 5$, we get $R_0 = 2$, confirming the analytic formula at the shared anchor.

R_0 for SEIR

Adding a non-infectious exposed stage E does not change R_0 , because E does not contribute to transmission. The transmissibility T depends only on the infectious stages.

```

@parameters σ

seir_prog = DiseaseProgression(
    [
        DiseaseStage(:E; transmission_rate=0),
        DiseaseStage(:I; transmission_rate=β),
        DiseaseStage(:R)
    ],
    [
        DiseaseTransition(:E, :I, σ),
        DiseaseTransition(:I, :R, γ)
    ];
    entry=:E
)
scm_seir = StaticConfigurationModel(pgf, seir_prog)
R0_seir = basic_reproduction_number(scm_seir)
println("Symbolic R₀ (SEIR): ", R0_seir)

```

Symbolic R_0 (SEIR): $(\beta \cdot \kappa) / (\beta + \gamma)$

```

R0_seir_val = Symbolics.value(
    Symbolics.substitute(R0_seir, Dict(β ⇒ β_anchor, γ ⇒ γ_anchor, σ ⇒ 0.2, κ ⇒ κ_anchor)
)
println("SEIR R0 = ", R0_seir_val)
println("SIR R0 = ", R0_val)
println("Equal? ", R0_seir_val ≈ R0_val)

```

```

SEIR R0 = 2.0000000000000004
SIR R0 = 2.0000000000000004
Equal? true

```

As expected, the SEIR R_0 equals the SIR R_0 because the latent period affects the epidemic *timing* but not the *threshold*.

Multi-stage R_0

For diseases with multiple infectious stages (e.g., acute and chronic phases with different transmission rates), the transmissibility becomes:

$$T = 1 - \prod_{m \in \text{infectious}} \frac{\gamma_m}{\beta_m + \gamma_m}$$

where the product is over all infectious stages. Each factor $\gamma_m/(\beta_m + \gamma_m)$ is the probability of progressing *without* transmitting in stage m , so T is the probability of transmitting in at least one stage.

```

@parameters β1 β2 γ1 γ2

twostage_prog = DiseaseProgression(
    [
        DiseaseStage(:I1; transmission_rate=β1),
        DiseaseStage(:I2; transmission_rate=β2),
        DiseaseStage(:R)
    ],
    [
        DiseaseTransition(:I1, :I2, γ1),
        DiseaseTransition(:I2, :R, γ2)
    ];
    entry=:I1
)

```

```
scm_twostage = StaticConfigurationModel(pgf, twostage_prog)
R0_twostage = basic_reproduction_number(scm_twostage)
println("Symbolic R0 (two-stage): ", R0_twostage)
```

Symbolic R_0 (two-stage): $(((\beta_1 + \gamma_1) * (\beta_2 + \gamma_2) - \gamma_1 * \gamma_2) * \kappa) / ((\beta_1 + \gamma_1) * (\beta_2 + \gamma_2))$

```
R0_twostage_val = Symbolics.value(
    Symbolics.substitute(R0_twostage,
        Dict{β1 ⇒ 0.3, γ1 ⇒ 0.2, β2 ⇒ 0.1, γ2 ⇒ 0.3, κ ⇒ 5.0})
)
expected_T = 1 - (0.2 / (0.3 + 0.2)) * (0.3 / (0.1 + 0.3))
expected_R0 = expected_T * 5.0
println("Computed R0 = ", R0_twostage_val)
println("Expected T = ", expected_T)
println("Expected R0 = ", expected_R0)
println("Match? ", R0_twostage_val ≈ expected_R0)
```

```
Computed R0 = 3.5
Expected T = 0.7
Expected R0 = 3.5
Match? true
```

Inspecting Model Equations

EdgeBasedModels.jl generates ODE systems using ModelingToolkit.jl. We can inspect these equations symbolically before solving.

Compact form

The compact (Miller) formulation uses just 2 ODEs for SIR:

```
compact_model = build_sir(pgf, β, γ; form=:compact)
compact_eqs = ModelingToolkit.equations(compact_model.system)
println("Compact SIR: ", length(compact_eqs), " ODEs")
for eq in compact_eqs
    println(" ", eq)
end
```

Compact SIR: 2 ODEs

```
Differential(t, 1)(R(t)) ~ (1 - R(t) - exp((-1 + θ(t))*κ)*(1 - ρ))*γ
Differential(t, 1)(θ(t)) ~ (θ(t)*exp(0)*β - (1 - θ(t))*exp(0)*γ - exp((-1 + θ(t))*κ)*β*(1 - ρ)) / (-exp(0))
```

Expanded form

The expanded formulation introduces explicit φ variables for each edge state:

```
expanded_model = build_sir(pgf, β, γ; form=:expanded)
expanded_eqs = ModelingToolkit.equations(expanded_model.system)
println("Expanded SIR: ", length(expanded_eqs), " ODEs")
for eq in expanded_eqs
    println(" ", eq)
end
```

Expanded SIR: 5 ODEs

```
Differential(t, 1)(pop_R(t)) ~ pop_I(t)*γ
Differential(t, 1)(pop_I(t)) ~ -pop_I(t)*γ + exp((-1 + θ(t))*κ)*phi_I(t)*β*κ*(1 - ρ)
Differential(t, 1)(phi_R(t)) ~ phi_I(t)*γ
Differential(t, 1)(phi_I(t)) ~ -phi_I(t)*(β + γ) + phi_S(t)*phi_I(t)*β*κ
Differential(t, 1)(θ(t)) ~ -phi_I(t)*β
```

We can also inspect the state variables and observables:

```
println("\nState variables (unknowns):")
for u in ModelingToolkit.unknowns(expanded_model.system)
    println(" ", u)
end
```

State variables (unknowns):

```
pop_R(t)
pop_I(t)
phi_R(t)
phi_I(t)
θ(t)
```

```
println("Observed (algebraic) variables:")
for obs in ModelingToolkit.observables(expanded_model.system)
    println(" ", obs)
end
```

Observed (algebraic) variables:

$$\begin{aligned}\phi_S(t) &\sim (\exp((-1 + \theta(t))\kappa) * (1 - \rho)) / \exp(\theta) \\ S(t) &\sim \exp((-1 + \theta(t))\kappa) * (1 - \rho) \\ I(t) &\sim \text{pop}_I(t)\end{aligned}$$

The expanded form makes the edge-level dynamics explicit: each ϕ variable tracks the probability that a random test edge connects to a neighbour in a given disease stage. The compact form combines these into a single θ equation using the PGF.

SEIR equations

For more complex diseases, the number of equations grows with the number of disease stages:

```
seir_model = build_seir(pgf, σ, β, γ)
seir_eqs = ModelingToolkit.equations(seir_model.system)
println("Expanded SEIR: ", length(seir_eqs), " ODEs")
for eq in seir_eqs
    println(" ", eq)
end
```

Expanded SEIR: 7 ODEs

$$\begin{aligned}\text{Differential}(t, 1)(\text{pop}_R(t)) &\sim \text{pop}_I(t) * \gamma \\ \text{Differential}(t, 1)(\text{pop}_I(t)) &\sim -\text{pop}_I(t) * \gamma + \text{pop}_E(t) * \sigma \\ \text{Differential}(t, 1)(\text{pop}_E(t)) &\sim -\text{pop}_E(t) * \sigma + \exp((-1 + \theta(t))\kappa) * \phi_I(t) * \beta * \kappa * (1 - \rho) \\ \text{Differential}(t, 1)(\phi_R(t)) &\sim \phi_I(t) * \gamma \\ \text{Differential}(t, 1)(\phi_I(t)) &\sim \phi_E(t) * \sigma - \phi_I(t) * (\beta + \gamma) \\ \text{Differential}(t, 1)(\phi_E(t)) &\sim -\phi_E(t) * \sigma + \phi_S(t) * \phi_I(t) * \beta * \kappa \\ \text{Differential}(t, 1)(\theta(t)) &\sim -\phi_I(t) * \beta\end{aligned}$$

Parameter Sensitivity of R_0

Because R_0 is a symbolic expression, we can evaluate it over parameter ranges to study sensitivity. This is much faster than sweeping parameters through ODE solutions.

R_0 versus mean degree κ


```

κ_range = 0.1:0.1:10.0
R0_vs_κ = [
    Symbolics.value(Symbolics.substitute(R0_sir, Dict{β ⇒ β_anchor, γ ⇒ γ_anchor, κ ⇒ κ_v}
    for κ_v in κ_range
]

plot(κ_range, R0_vs_κ, lw=2, color=:steelblue, label="R0(κ)")
hline!([1.0], linestyle=:dash, color=:red, label="Epidemic threshold")
xlabel!("Mean degree κ")
ylabel!("R0")
title!("R0 vs Network Connectivity")

```

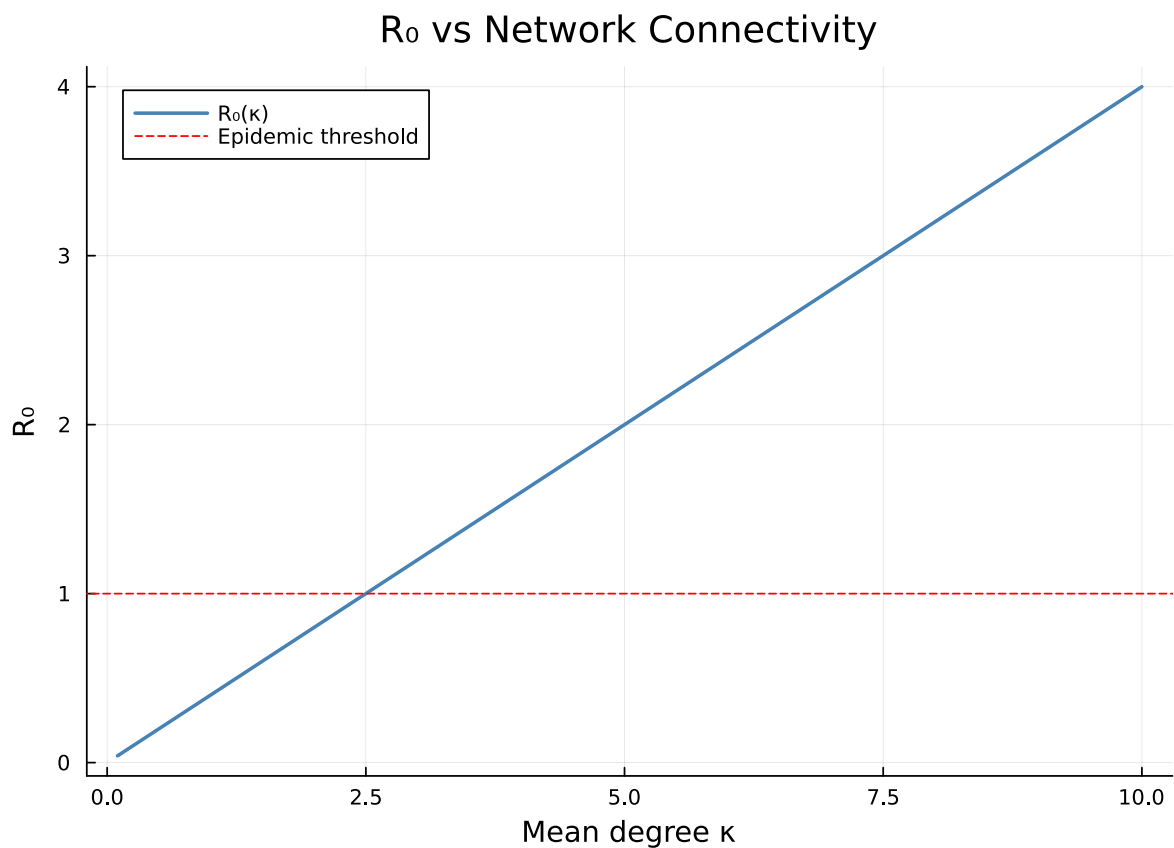


Figure 1: Figure 1: R_0 as a function of mean network degree κ , for anchored $\beta = 1/6$ and $\gamma = 0.25$. The dashed line marks the epidemic threshold $R_0 = 1$.

For a Poisson network with $R_0 = \beta\kappa/(\beta+\gamma)$, the epidemic threshold $R_0 = 1$ occurs at $\kappa^* = (\beta+\gamma)/\beta$. With the anchored values $\beta = 1/6$ and $\gamma = 0.25$, this gives $\kappa^* = 2.5$.

R_0 versus transmission rate β

```
β_range = 0.01:0.01:1.0
R0_vs_β = [
    Symbolics.value(Symbolics.substitute(R0_sir, Dict(β ⇒ β_v, γ ⇒ γ_anchor, κ ⇒ κ_anchor)
    for β_v in β_range
]

plot(β_range, R0_vs_β, lw=2, color=:darkorange, label="R0(β)")
hline!([1.0], linestyle=:dash, color=:red, label="Epidemic threshold")
xlabel!("Transmission rate β")
ylabel!("R0")
title!("R0 vs Transmission Rate")
```

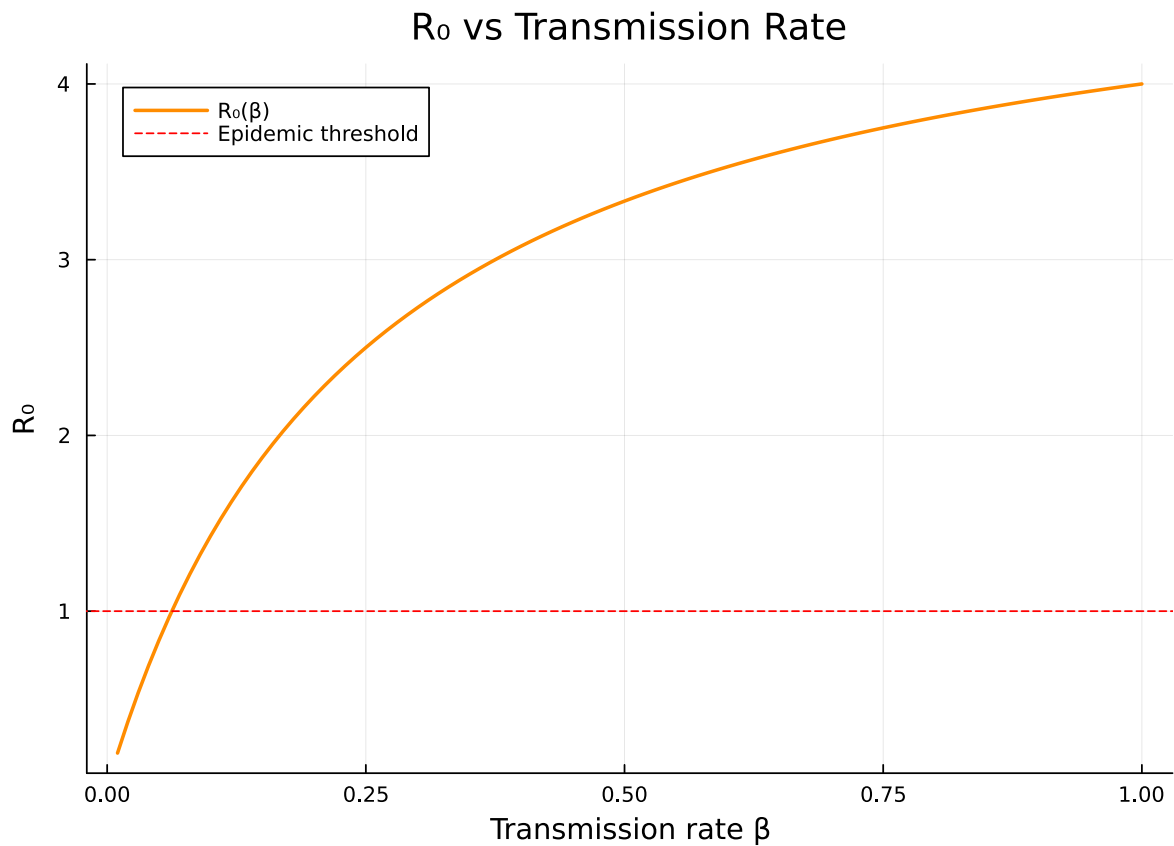


Figure 2: Figure 2: R_0 as a function of transmission rate β , for fixed $\kappa = 5$ and anchored $\gamma = 0.25$.

R_0 increases with β but saturates at $\kappa = 5$ as $\beta \rightarrow \infty$ (since $\beta/(\beta + \gamma) \rightarrow 1$).

Multi-stage R_0 sensitivity

We can also examine how R_0 depends on the relative transmission rates in a two-stage model:

```
β1_range = 0.01:0.02:0.5
β2_range = 0.01:0.02:0.5
R0_matrix = [
    Symbolics.value(Symbolics.substitute(R0_twostage,
        Dict{β1 ⇒ b1, β2 ⇒ b2, γ1 ⇒ 0.2, γ2 ⇒ 0.3, κ ⇒ 5.0}))
    for b1 in β1_range, b2 in β2_range
]

heatmap(collect(β1_range), collect(β2_range), R0_matrix',
    xlabel="β1 (stage 1 transmission)",
    ylabel="β2 (stage 2 transmission)",
    title="Two-stage R0",
    colorbar_title="R0",
    color=:viridis)
contour!(collect(β1_range), collect(β2_range), R0_matrix',
    levels=[1.0], linewidth=2, linecolor=:white, label="")
```

PGF Analysis

The probability generating function (PGF) encodes the network's degree distribution and plays a central role in edge-based models. `EdgeBasedModels.jl` provides tools for symbolic manipulation of PGFs.

Poisson PGF derivatives

For a Poisson network with mean degree κ , the PGF is $\psi(z) = e^{\kappa(z-1)}$:

```
pgf_poisson = poisson_pgf(κ)
println("ψ(z) = ", pgf_poisson.expression)
```

$$\psi(z) = \exp((-1 + z) \cdot \kappa)$$

The first and second derivatives are:

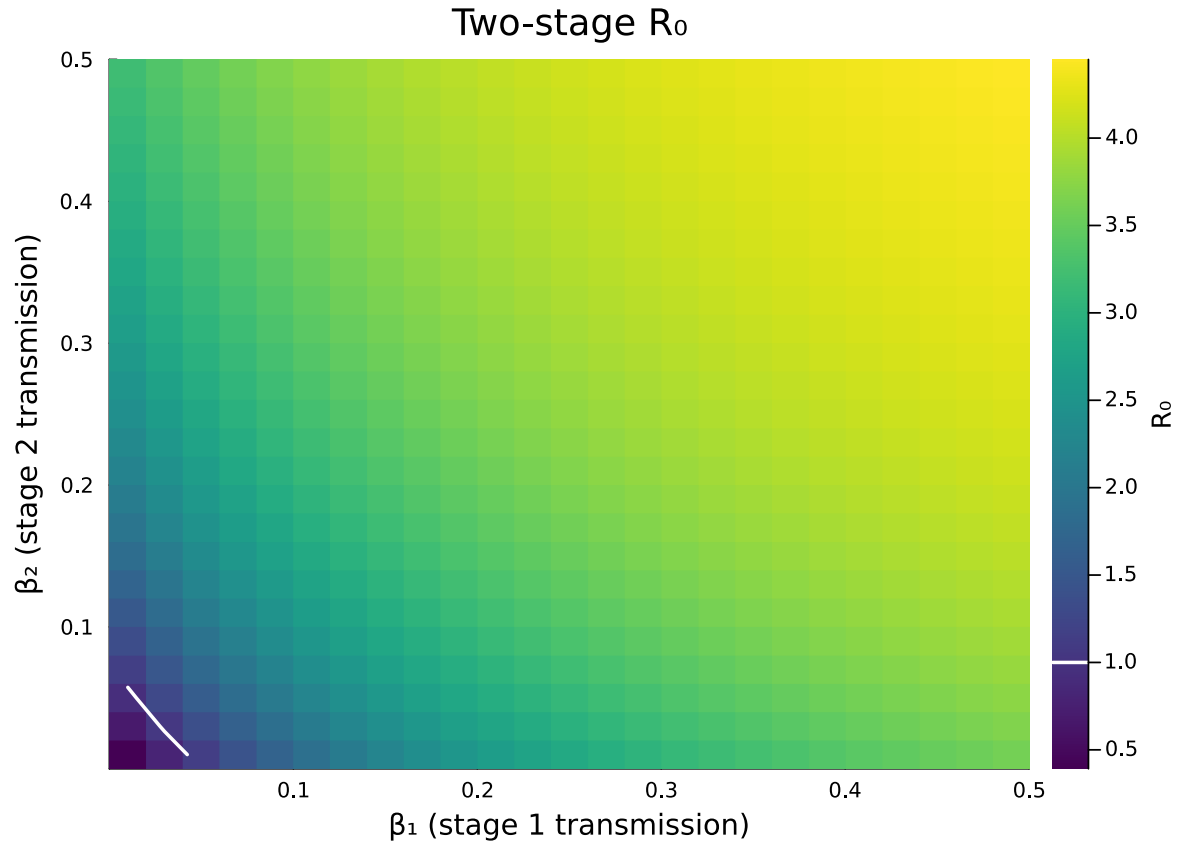


Figure 3: Figure 3: Two-stage R_0 as a function of β_1 and β_2 , with $\gamma_1 = 0.2$, $\gamma_2 = 0.3$, $\kappa = 5$. The white contour marks $R_0 = 1$.

```
d1 = pgf_derivative(pgf_poisson, 1)
println("ψ'(z) = ", d1)

d2 = pgf_derivative(pgf_poisson, 2)
println("ψ''(z) = ", d2)
```

$\psi'(z) = \exp((-1 + z) \cdot \kappa) \cdot \kappa$
 $\psi''(z) = \exp((-1 + z) \cdot \kappa) \cdot (\kappa^2)$

The mean degree is $\psi'(1)$ and the excess degree ratio is $\psi''(1)/\psi'(1)$:

```
mean_deg = mean_degree(pgf_poisson)
println("Mean degree ψ'(1) = ", mean_deg)

excess = Symbolics.simplify(
    Symbolics.substitute(d2, Dict{pgf_poisson.variable ⇒ 1}) /
    Symbolics.substitute(d1, Dict{pgf_poisson.variable ⇒ 1})
)
println("Excess degree ψ''(1)/ψ'(1) = ", excess)
```

Mean degree $\psi'(1) = \exp(0) \cdot \kappa$
Excess degree $\psi''(1)/\psi'(1) = \kappa$

For a Poisson distribution, the excess degree equals the mean degree — a special property that simplifies the R_0 expression.

Polynomial PGF

For non-Poisson networks, the excess degree differs from the mean degree. Consider a network with degree distribution $P(k=0) = 0.2$, $P(k=1) = 0.3$, $P(k=2) = 0.5$:

```
pgf_poly = polynomial_pgf([0.2, 0.3, 0.5])
println("ψ(z) = ", pgf_poly.expression)
println("Mean degree: ", Symbolics.value(mean_degree(pgf_poly)))
```

$\psi(z) = 0.2 + 0.3z + 0.5(z^2)$
Mean degree: 1.3

```

d1_poly = pgf_derivative(pgf_poly, 1)
d2_poly = pgf_derivative(pgf_poly, 2)

d1_at_1 = Symbolics.value(Symbolics.substitute(d1_poly, Dict(pgf_poly.variable => 1)))
d2_at_1 = Symbolics.value(Symbolics.substitute(d2_poly, Dict(pgf_poly.variable => 1)))

println("ψ'(1) = ", d1_at_1, " (mean degree)")
println("ψ''(1) = ", d2_at_1)
println("Excess degree ratio ψ''(1)/ψ'(1) = ", d2_at_1 / d1_at_1)

```

```

ψ'(1) = 1.3 (mean degree)
ψ''(1) = 1
Excess degree ratio ψ''(1)/ψ'(1) = 0.7692307692307692

```

For this distribution, the mean degree is 1.3 but the excess degree ratio is $1.0/1.3 \approx 0.77$ — neighbours tend to have *fewer* additional contacts than the mean degree suggests. This is because the degree distribution is concentrated on low values ($k \leq 2$).

Comparing R_0 across degree distributions

We can use the symbolic framework to compare R_0 for different network structures with the same mean degree:

```

@parameters β_plot γ_plot

# Poisson  $R_0$  for varying mean degree
κ_vals = 1.0:0.1:8.0
R0_poisson = [
    Symbolics.value(Symbolics.substitute(R0_sir, Dict(β => 0.3, γ => 0.1, κ => k)))
    for k in κ_vals
]

# Polynomial PGF  $R_0$ : for each mean degree, construct a degree distribution
# with  $P(0) = 1 - \mu/k_{\max}$ , ..., concentrated on  $\{0, k_{\max}\}$ 
# Using a simple two-point distribution:  $P(0) = 1 - \mu/k_{\max}$ ,  $P(k_{\max}) = \mu/k_{\max}$ 
k_max = 10
R0_poly = Float64[]
for μ in κ_vals
    if μ > k_max
        push!(R0_poly, NaN)
    end
end

```

```

        continue
    end
    probs = zeros(k_max + 1)
    probs[1] = 1.0 - μ / k_max      # P(k=0)
    probs[k_max + 1] = μ / k_max    # P(k=k_max)
    pgf_p = polynomial_pgf(probs)
    prog_p = DiseaseProgression(
        [DiseaseStage(:I; transmission_rate=β_plot), DiseaseStage(:R)],
        [DiseaseTransition(:I, :R, γ_plot)]; entry=:I
    )
    R0_p = basic_reproduction_number(StaticConfigurationModel(pgf_p, prog_p))
    push!(R0_poly, Symbolics.value(Symbolics.substitute(R0_p, Dict{β_plot => 0.3, γ_plot => 0.3})))
end

plot(κ_vals, R0_poisson, lw=2, label="Poisson", color=:steelblue)
plot!(κ_vals, R0_poly, lw=2, label="Two-point (0, $k_max)", color=:darkorange)
hline!([1.0], linestyle=:dash, color=:red, label="Epidemic threshold")
xlabel!("Mean degree")
ylabel!("R₀")
title!("R₀ by Degree Distribution")

```

Summary

The symbolic nature of `EdgeBasedModels.jl` enables powerful analytical workflows:

- R_0 computation: Exact symbolic expressions for the basic reproduction number, valid for arbitrary disease progressions and degree distributions.
- Equation inspection: Full visibility into the ODE systems before solving, useful for verification and teaching.
- Parameter sensitivity: Fast evaluation of R_0 across parameter ranges without running ODE solvers.
- PGF analysis: Symbolic derivatives of the degree distribution's PGF reveal how network structure affects epidemic thresholds.

These capabilities complement the numerical simulation workflow shown in earlier vignettes, providing analytical insight that guides parameter choices and model validation.

Simulation validation

This vignette focuses on a scenario for which `NetworkOutbreaks.jl` does not yet provide an out-of-the-box stochastic ground truth (multi-type host graphs with prescribed mixing matrices, time-

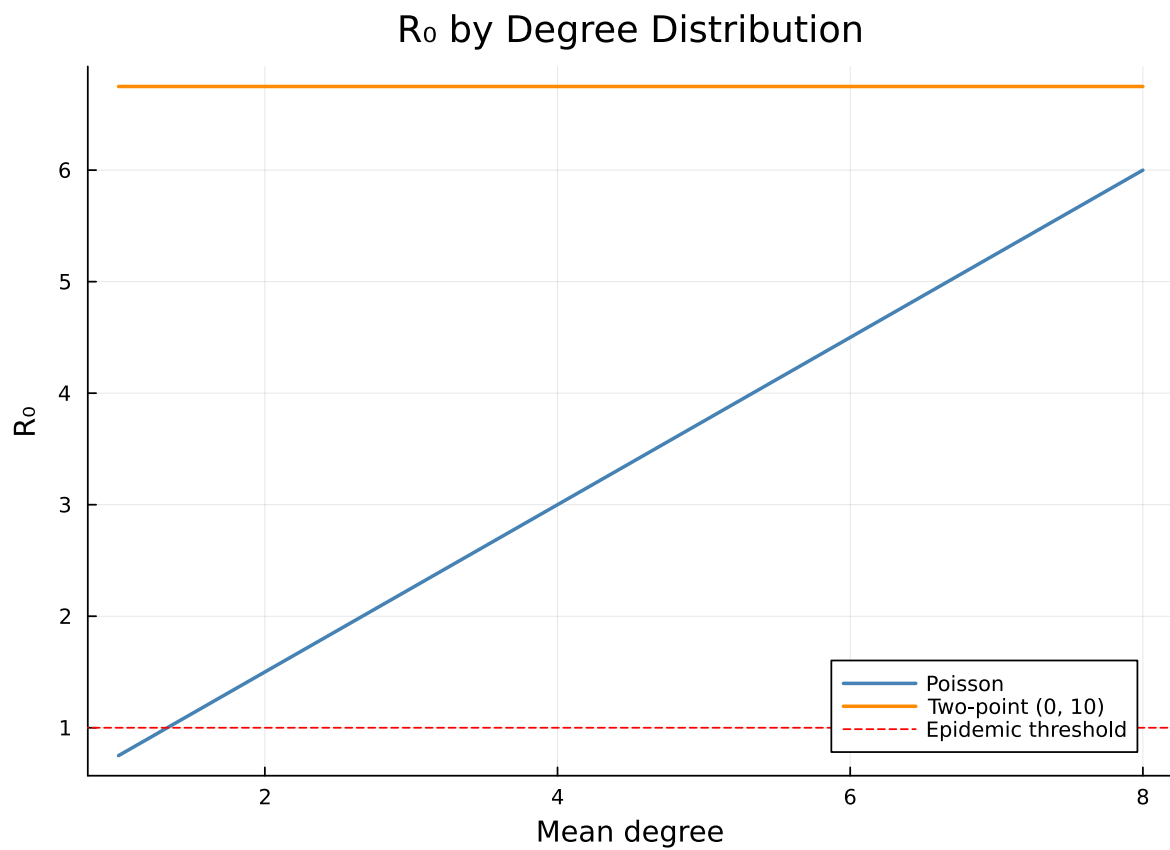


Figure 4: Figure 4: R_0 for Poisson and polynomial degree distributions, both with the same mean degree. The Poisson network always has higher R_0 because its excess degree equals its mean degree.

varying networks via the EBCM rewiring schedule, multiplex layers, degree-correlated configuration models, clustered graphs with prescribed triangle counts, or final-size sweeps over R_0).

A future revision will add the missing primitives to `NetworkOutbreaks.jl` and overlay the corresponding Gillespie SSA ribbon here. Until then, see [vignette 01](#) for the validation pattern on a single-layer Poisson configuration-model network with the same canonical parameters.